

Department of Information Engineering and Computer Science

Bachelor's Degree in
Computer Science

Bachelor's Thesis

Galactic Tidal Stripping

Numerical Simulation and Prediction of Star Cluster Evolution

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Academic Year 2026/2027

Acknowledgments

A heartfelt thank you to…

To my father, for his practical support, including financial assistance, which allowed me to pursue this wonderful journey.

To my mother, for listening to me, giving me strength during the most difficult moments, and never allowing me to give up.

To my brother, who introduced me to the world of computer science and inspired in me a passion for this discipline.

To all my family, for the support and trust they have shown me over the years.
To those who stand by me and cheer me on every day, and to those who are not physically here today but continue to accompany me in my thoughts and live forever in my heart.

To all my friends, with whom I have shared laughter, carefree moments, and even less peaceful days. Your presence has made this journey easier and turned these years into a chapter of my life that I will always hold dear.

To my co-supervisors, for their availability, patience, and valuable advice, but above all for embracing my ideas and guiding me in turning my passions into something tangible.

Finally, to myself. For finding the strength to keep going even when everything seemed impossible, for not giving up during the darkest moments, and for making it this far.
May this achievement serve as proof, first and foremost to myself, that one must never stop believing, even when giving up may seem like the easiest choice.

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"The Galaxy is nothing else but a countless number of stars scattered in clusters." Galileo Galilei, *Sidereus Nuncius*, 1610

Abstract

Star clusters are gravitationally bound systems whose evolution can be profoundly influenced by the galactic environment in which they are located. In particular, interaction with the gravitational field of the galaxy can cause a progressive loss of stars, a phenomenon known as galactic tidal stripping, which in the most extreme cases can lead to the complete dissolution of the cluster.
Studying these processes requires numerical simulations of many-body systems, whose computational cost can become significant when numerous initial configurations need to be analyzed. This motivates the use of machine learning techniques, which can derive useful relationships from simulation results to make new predictions.

The aim of this work is to study how initial orbital parameters influence the survival of star clusters and to use simulation results to train models capable of predicting their evolutionary outcomes. In particular, the orbital parameters considered are the semi-major axis and eccentricity of the cluster's orbit. The Plummer model was adopted to describe the system, both to generate the initial distribution of stellar positions and velocities and to represent the galactic gravitational potential. The velocities were then rescaled to obtain configurations close to virial equilibrium. The defined equations of motion include gravitational interactions between the stars and the influence of the galactic field. Procedures were also developed to calculate diagnostic quantities, including the bound mass fraction and the half-mass radius, which are necessary to describe the state of the clusters at the end of the simulations.

To construct the *dataset*, 1140 orbital configurations were considered, obtained by combining 30 values of the semi-major axis and 38 values of the eccentricity. Each configuration was simulated using 15 different initial realizations of the cluster, for a total of 17100 simulations. Analysis of the results made it possible to observe the relationship between orbital parameters and cluster mass loss, showing that survival depends on both the semi-major axis and eccentricity.
The collected data were subsequently used to formulate a binary classification problem, distinguishing surviving clusters from dissolved ones based on their final diagnostic quantities. The semi-major axis and eccentricity were selected as the input variables for the learning models, while model performance was evaluated using the *F1-score*, with the *dataset* divided into a training set and a test set.

Two classifiers were implemented and compared: a *Support Vector Classifier* and a *Multi-Layer Perceptron* neural network. Both achieved high and very similar performance. The decision boundaries identified by the two models closely follow the separation between survival and dissolution observed in the distribution of simulation results. The main classification difficulties are concentrated in the transition region between the two regions, where clusters with the same orbital parameters can reach different outcomes depending on their initial realization.

The results show that machine learning techniques can effectively describe the relationship between initial orbital conditions and the final state of star clusters, offering a possible alternative to numerical simulations for making predictions about new configurations. The work also highlights several directions for future development, including extending the range of orbital parameters studied, increasing the number of stars and initial realizations, and assessing the models' ability to predict outcomes for configurations not included in the training set.

Chapter 1

Introduction

Looking up at a clear night sky, we are faced with one of the most majestic spectacles nature has to offer: the starry sky.



Figure 1.1: The starry sky above the Paranal Observatory in Chile (Credits: R. Wesson/ESO).

The starry sky presents an image of a seemingly motionless universe, in which stars appear as fixed points of light, separated by distances that are difficult to imagine. In reality, what we observe is merely a snapshot of a constantly evolving cosmos. In interstellar space, immense concentrations of gas and dust, known as nebulae, can collapse under the influence of gravity, giving birth to new stars. On even larger scales, the same force helps organise matter into increasingly complex structures, from planetary systems to entire galaxies.



Figure 1.2: The Orion Nebula (left, credits: ESO/G. Beccari) and the Carina Nebula (right, credits: ESO; VPHAS+ Consortium/Cambridge Astronomical Survey Unit).

1.1 Stellar Clusters

When numerous stars are born in the same region of space, they can remain gravitationally bound to one another, giving rise to self-gravitating systems known as stellar clusters.
In these systems, each star contributes to the overall gravitational field and is continuously affected by the attraction exerted by all the other components [18].

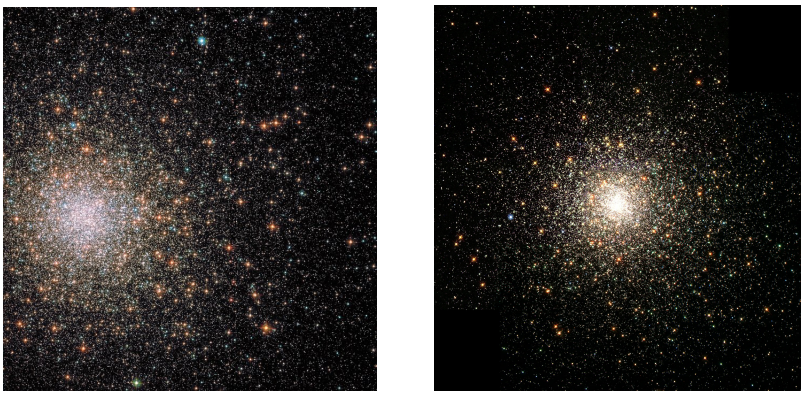


Figure 1.3: The globular clusters Messier 62 (left, credits: ESA/Hubble) and Messier 80 (right, credits: NASA).

These magnificent and imposing structures are particularly sensitive to gravitational interactions with their surrounding environment. The influence exerted by the galaxy that hosts them can profoundly alter their structure and compromise their delicate gravitational equilibrium.

1.2 Galactic Tidal Stripping

As a stellar cluster orbits a galaxy, each of its stars experiences the galaxy's gravitational attraction to a different extent, depending on its position. These differences in the strength of gravitational attraction generate the so-called tidal forces, which can profoundly alter the cluster's equilibrium and evolution. The progressive loss of stars caused by this phenomenon is known as galactic tidal stripping and, in the most extreme cases, can lead to the complete disruption of the cluster, accompanied by the formation of long stellar structures that follow the orbit of the original system, known as tidal tails [2].

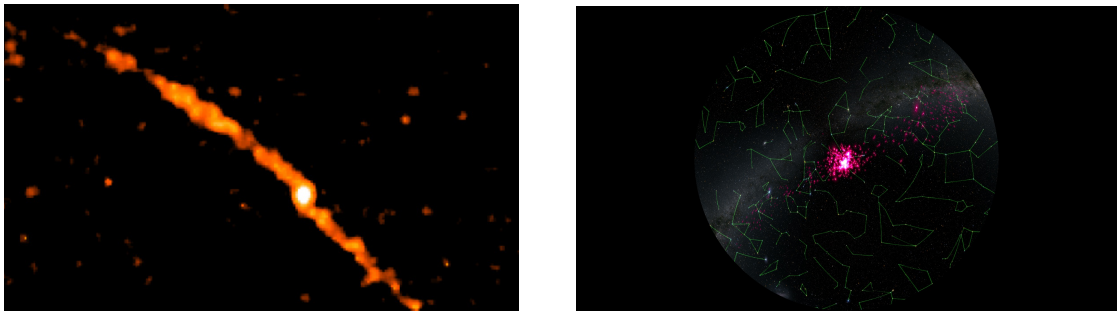


Figure 1.4: Tidal tails of the globular cluster Palomar 5 (left, credits: SDSS) and a representation of the tidal tails of the Hyades open cluster, highlighted in pink (right, credits: ESA/Gaia/DPAC; S. Jordan/T. Sagrista).

Understanding the nature and intensity of this phenomenon is essential for studying the evolution and survival of stellar clusters. However, these processes unfold over extremely long timescales, which can span tens of millions of years.

1.3 Computational Astrophysics

Many astronomical phenomena unfold over timescales that cannot be directly followed through observations. To study them, modern astrophysics increasingly relies on mathematical models and computational tools. This approach, known as computational astrophysics, makes it possible to numerically reconstruct complex systems and analyse their evolution. A significant example is the *Millennium Simulation*, which made it possible to study the formation of cosmic structures by tracking billions of dark matter particles [30].

1.4 Numerical Simulations

To describe the evolution of a stellar cluster, each star can be represented as a particle with mass, position and velocity, and its motion can be calculated using the laws of gravity. This approach is known as an N-body simulation and makes it possible to numerically follow the interactions between all components of the system, also accounting for the influence of an external galactic gravitational force. The main difficulty lies in the computational cost: in the direct summation method, the number of gravitational interactions to be calculated grows proportionally to the square of the number of stars in the system, quickly making the simulation of large systems computationally expensive. Simulating numerous initial configurations can therefore require considerable resources and time [31].
This need has led to an interest in tools capable of using simulation results to make new predictions in a much shorter time.

1.5 Machine Learning

Machine learning is a set of computational techniques that allows algorithms to learn relationships and recurring patterns from data. In astrophysics, it is used, among other applications, to classify celestial objects, analyse observations and build predictive models [25]. When applied to the results of numerical simulations, it makes it possible to learn the relationship between a system's initial conditions and its final behaviour. In this way, a trained model can provide estimates for new configurations without necessarily having to repeat the entire simulation each time, offering a complementary tool to traditional numerical methods.

1.6 Objective

To study the evolution of stellar clusters subject to galactic tidal stripping, analysing how their initial orbital parameters influence their survival. To use the data obtained from simulations to train machine learning models, with the aim of predicting the final state of clusters from their initial orbital conditions and thereby reducing the computational cost required to obtain these estimates.

Chapter 2

The Physical Model

The physical model adopted describes stellar clusters as systems of point-like stars, subject to their mutual gravitational attraction and to the external field of the galaxy. The Plummer model is used to represent the initial distribution of stars, from which their initial positions and velocities are generated. The virial equilibrium of the cluster is then checked, and the velocities are rescaled to obtain an initial configuration closer to equilibrium. The galaxy is also described using a Plummer potential. The equations of motion are then defined, accounting for both the gravitational interactions between the stars and the influence of the galaxy, and the initial orbital conditions are established through the semi-major axis and eccentricity. Once the initial conditions have been defined, the equations of motion are numerically integrated and energy conservation is monitored throughout the evolution of the system. Finally, to evaluate the effects of tidal stripping, the stars that remain gravitationally bound to the cluster are identified, and the centre of mass, bound mass fraction and half-mass radius are calculated. These quantities make it possible to describe the evolution of the cluster and quantify its mass loss and structural changes.

2.1 Notation

Table 2.1 summarises the main symbols used throughout this document.

Table 2.1: Main symbols used throughout this document.

Symbol	Meaning	Section
N	Number of stars in the cluster	2.2.1
m	Mass of a single star	2.2.1
M_{tot}	Total mass of the cluster	2.2.1
A	Scale parameter of the Plummer model of the cluster	2.2.1
r	Radial distance from the centre of the stellar cluster	2.2.1
ρ	Volume mass density of the cluster	2.2.1
Φ	Gravitational potential of the cluster	2.2.1
G	Gravitational constant	2.2.1
$\mathbf{x}_i$	Position vector of star i	2.2.2
$\mathbf{v}_i$	Velocity vector of star i	2.2.3
M_{gal}	Mass of the galactic model	2.3.1
A_{gal}	Scale parameter of the galactic Plummer model	2.3.1
R	Distance from the galactic centre	2.3.1
ϵ	Gravitational softening parameter	2.3.2
a	Semi-major axis of the orbit	2.3.3
e	Orbital eccentricity	2.3.3
Δt	Integration time step	2.3.4
$\mathbf{x}_{\text{CM,bound}}$	Position of the centre of mass of the bound stars	2.3.6
f_{bound}	Bound mass fraction	2.3.7
$r_{1/2}$	Half-mass radius	2.3.8

2.2 Stellar Cluster Generation

To describe the initial distribution of a stellar cluster, the Plummer model [22] is adopted, allowing the distribution of stellar positions and velocities within the galactic context to be defined.

2.2.1 The Plummer Model

The Plummer model describes a self-gravitating spherical system through its mass distribution. It is characterised by the total mass M_{tot} and the scale parameter A, which determines the extent of the distribution. In the system considered, the cluster consists of N stars, each with mass m. The total mass is therefore given by the sum of the masses of the individual stars:

$$M_{\text{tot}} = Nm \tag{2.1}$$

The scale parameter A, on the other hand, defines the characteristic size of the cluster: for the same total mass, larger values correspond to a more extended and less concentrated distribution, whereas smaller values describe a more compact system. Following Plummer [22], the volume mass density as a function of the radial distance r from the centre is:

$$\rho(r) = \frac{3M_{\text{tot}}}{4\pi A^3}\left(1 + \frac{r^2}{A^2}\right)^{-5/2} \tag{2.2}$$

Binney and Tremaine [3] show that, by applying Poisson's equation to the mass distribution described in Equation (2.2), the following gravitational potential is obtained:

$$\Phi(r) = -\frac{GM_{\text{tot}}}{\sqrt{r^2 + A^2}} \tag{2.3}$$

2.2.2 Position Distribution

The initial positions of the stars are sampled so as to reproduce the spatial distribution described by the Plummer model. Letting r_i denote the distance of star i from the cluster centre, integrating the density defined in Equation (2.2) up to this distance gives the enclosed mass fraction:

$$\frac{M(< r_i)}{M_{\text{tot}}} = \frac{r_i^3}{(r_i^2 + A^2)^{3/2}} \tag{2.4}$$

This quantity represents the cumulative probability of finding a star at a distance from the centre less than or equal to r_i. By drawing a uniform random number u in the interval $(0, 1)$ and setting it equal to this probability, the initial distance of star i can be obtained:

$$r_i = \frac{A}{\sqrt{u^{-2/3} - 1}} \tag{2.5}$$

Once the distance r_i has been determined, the position direction is sampled isotropically. The position vector of star i in Cartesian coordinates is:

$$\mathbf{x}_i = \begin{pmatrix} r_i \sin\theta_i \cos\phi_i \\ r_i \sin\theta_i \sin\phi_i \\ r_i \cos\theta_i \end{pmatrix} \tag{2.6}$$

where ϕ_i is sampled uniformly in $[0, 2\pi)$ and $\cos\theta_i$ in $[-1, 1]$, thus obtaining a uniform distribution of directions in space. Repeating this procedure for all N stars produces a three-dimensional distribution consistent with the Plummer profile.

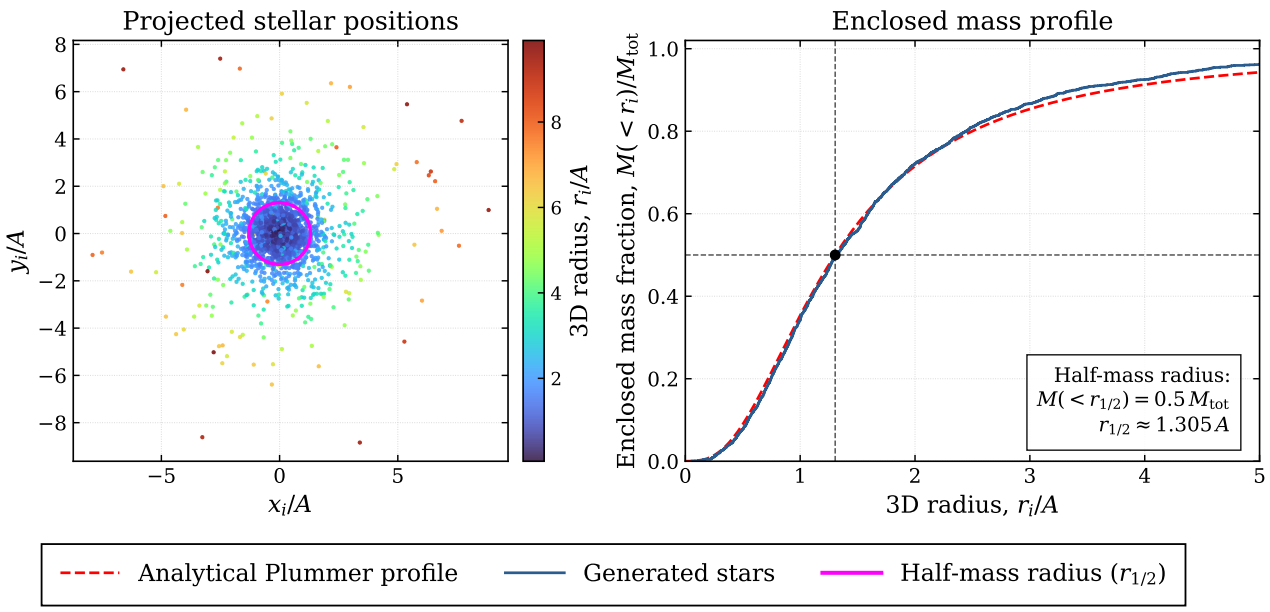


Figure 2.1: Three-dimensional distribution of the initial positions of stars generated according to the Plummer model. ($N = 2000$, $A = 1.0$)

2.2.3 Velocity Distribution

Once the initial positions have been determined, the velocities of individual stars are generated as a function of their distance r_i from the cluster centre. For each star, the local escape velocity is first calculated:

$$v_{\rm esc}(r_i) = \sqrt{-2\Phi(r_i)} = \sqrt{\frac{2GM_{\rm tot}}{\sqrt{r_i^2 + A^2}}} \tag{2.7}$$

The velocity magnitude is sampled according to the theoretical distribution predicted by the isotropic Plummer model:

$$p(v_i \mid r_i) \propto v_i^2 \left(1 - \frac{v_i^2}{v_{\rm esc}^2(r_i)}\right)^{7/2} \qquad 0 \le v_i \le v_{\rm esc}(r_i) \tag{2.8}$$

This distribution ensures that the initial velocities are lower than the local escape velocity, preventing stars that are initially unbound from the cluster from being generated. To sample from this distribution, the rejection sampling method [7] is used: a candidate velocity magnitude is generated in the interval $[0, v_{\rm esc}(r_i)]$ and accepted with a probability proportional to the theoretical distribution. Otherwise, the sample is rejected and generated again. The velocity magnitude of star i therefore corresponds to the accepted candidate value. Once the magnitude has been determined, the velocity direction is sampled isotropically in three-dimensional space. The velocity vector of star i is obtained by multiplying the magnitude by a randomly sampled unit direction vector $\hat{\mathbf{n}}_i$:

$$\mathbf{v}_i = v_i \hat{\mathbf{n}}_i \tag{2.9}$$

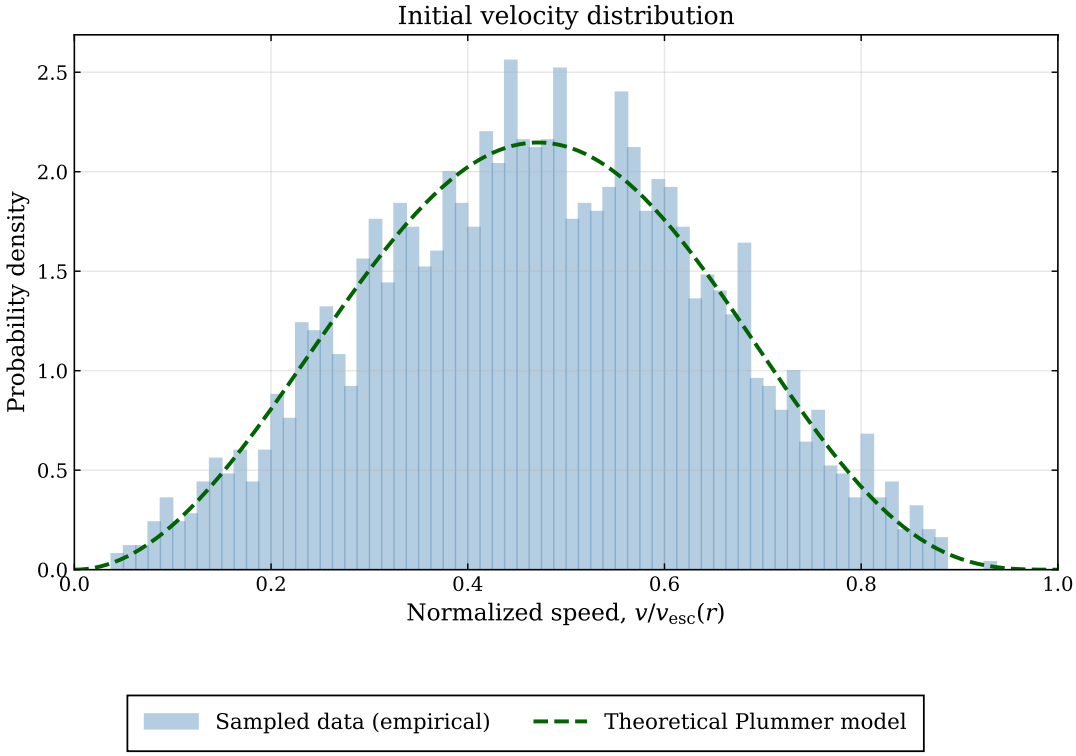


Figure 2.2: Distributions of velocity magnitudes normalised by the local escape velocity. ($N = 2000$, $A = 1.0$)

2.2.4 Virial Equilibrium Condition

To evaluate the initial dynamical equilibrium of the cluster, the total kinetic energy K and the gravitational potential energy U are considered [5]. The kinetic energy is given by the sum of the contributions from all stars:

$$K = \sum_{i=1}^{N} \frac{1}{2} m_i \|v_i\|^2 \tag{2.10}$$

The gravitational potential energy is obtained by summing the interactions between all pairs of stars:

$$U = -\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{G m_i m_j}{\|\mathbf{x}_i - \mathbf{x}_j\|} \tag{2.11}$$

For a self-gravitating system in virial equilibrium, the two energies satisfy the relation:

$$2K + U = 0 \tag{2.12}$$

The virial ratio makes it possible to quantify the deviation from this condition:

$$Q = \frac{2K}{|U|} \tag{2.13}$$

The value $Q = 1$ corresponds to virial equilibrium.

2.2.5 Velocity Rescaling

In the continuous Plummer model, the theoretical velocity distribution is consistent with the virial equilibrium condition described in Section 2.2.4, which is satisfied in the limit of an infinite number of stars ($N \to \infty$). In simulations, however, the number of stars is finite, and random sampling can result in a kinetic energy that does not exactly satisfy this condition.

To correct this deviation, the velocities are multiplied by a common factor λ, while the positions of the stars remain unchanged. This procedure does not alter the gravitational potential energy, which depends exclusively on the positions of the particles. Since kinetic energy is proportional to the square of the velocities, after rescaling we obtain:

$$K' = \lambda^2 K \tag{2.14}$$

By imposing that the final virial ratio is $Q' = 1$, the rescaling factor can be derived from the energies calculated before the correction:

$$\lambda = \sqrt{\frac{|U|}{2K}} = \frac{1}{\sqrt{Q}} \tag{2.15}$$

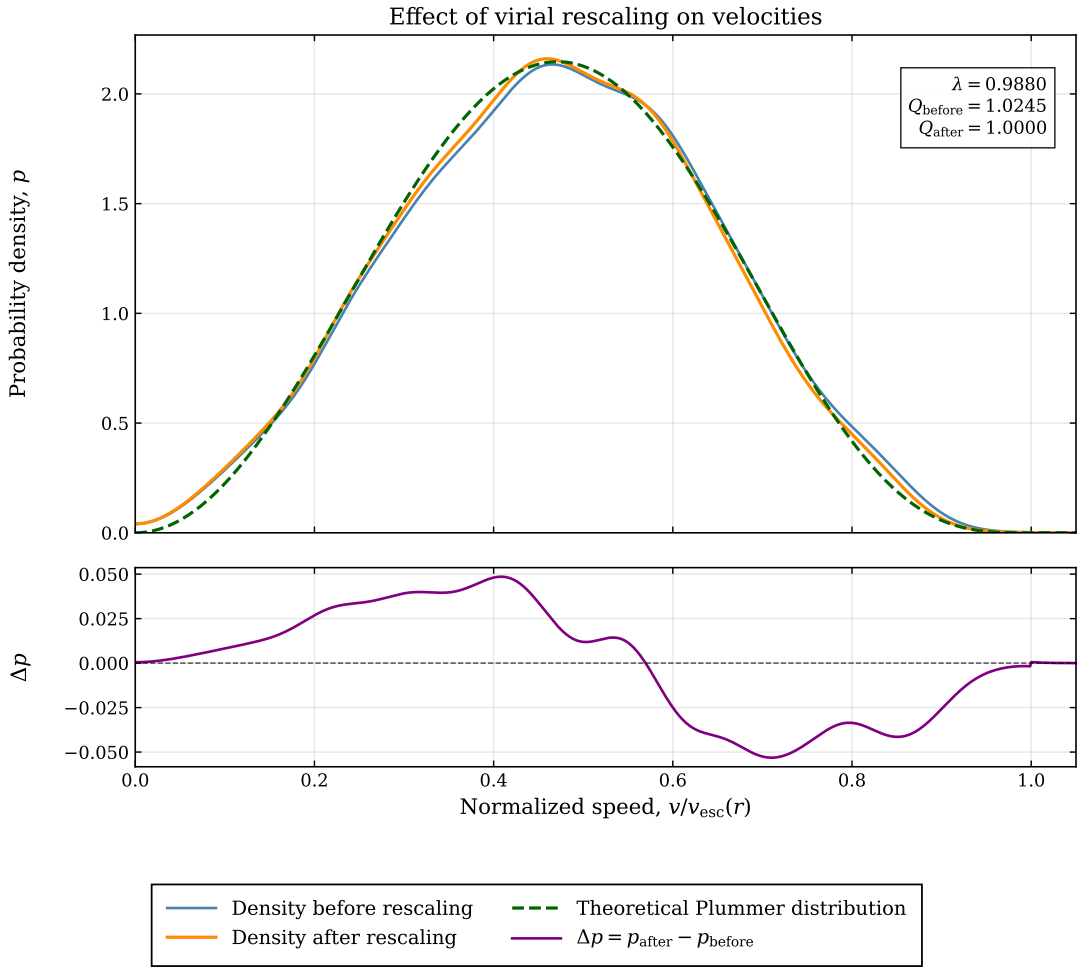


Figure 2.3: Distributions of velocity magnitudes normalised before and after rescaling. ($N = 2000$, $A = 1.0$)

2.3 Evolution and Orbital Dynamics

The evolution of the stellar cluster is described through the gravitational interaction between the stars and the external galactic potential, which determines its orbital dynamics and evolution over time.

2.3.1 Galactic Plummer Potential

The Plummer model [22], already introduced in Section 2.2.1 to model the mass distribution of the stellar cluster, is also adopted to describe the gravitational potential of the galaxy. In this case, the model represents the external gravitational field in which the cluster evolves and is expressed as a

function of the distance R from the galactic centre:

$$\Phi_{\text{gal}}(R) = -\frac{GM_{\text{gal}}}{\sqrt{R^2 + A_{\text{gal}}^2}} \tag{2.16}$$

The parameter A_{gal} represents the scale length of the galactic model and characterises its extent. The adopted form avoids the singularity of the potential at the centre, which occurs in the case of a point mass.
Starting from the potential defined in Equation (2.16), the gravitational acceleration exerted by the galaxy is obtained by calculating the negative gradient of the potential:

$$\mathbf{a}_{\text{gal}}(\mathbf{R}) = -\frac{GM_{\text{gal}}}{(R^2 + A_{\text{gal}}^2)^{3/2}}\mathbf{R} \tag{2.17}$$

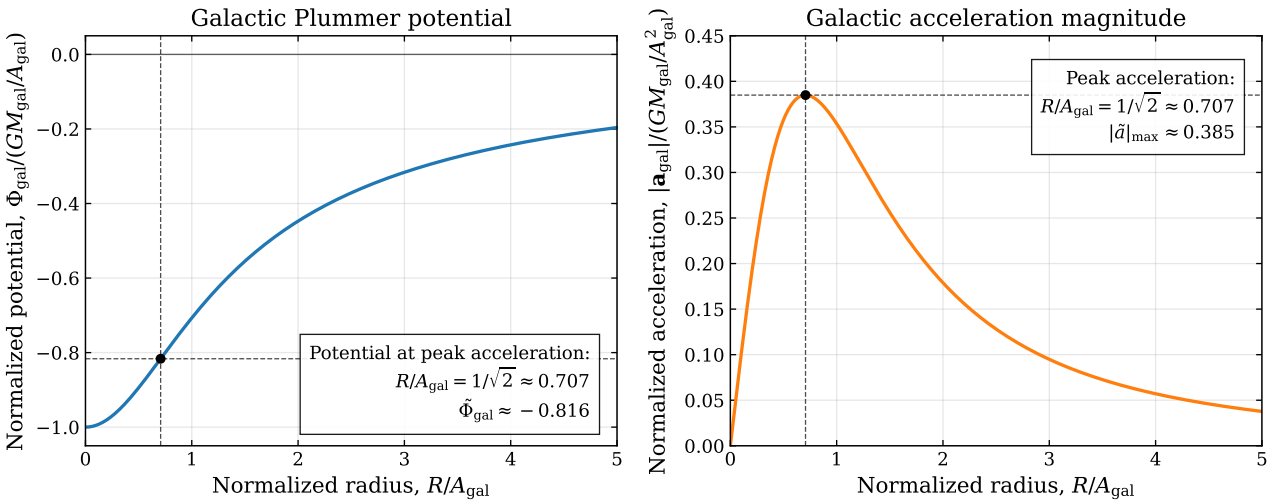


Figure 2.4: Gravitational potential and acceleration magnitude of the galactic Plummer model. ($G = 1.0$, $M_{\text{gal}} = 2000.0$, $A_{\text{gal}} = 10.0$)

2.3.2 Equations of Motion

The evolution of the stellar cluster is described by Newton's equations for a system of gravitationally interacting particles [9]. Letting $\mathbf{x}_i$ denote the position of star i and m_i its mass, the equations of motion are:

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i \qquad \frac{d\mathbf{v}_i}{dt} = \mathbf{a}_i \tag{2.18}$$

To regularise the gravitational potential during close encounters between stars, a softening parameter ϵ is introduced. The potential energy of interaction between two stars i and j therefore becomes:

$$U_{ij} = -\frac{Gm_i m_j}{\sqrt{\|\mathbf{x}_i - \mathbf{x}_j\|^2 + \epsilon^2}} \tag{2.19}$$

The internal contribution, accounting for the softening parameter, is expressed as:

$$\mathbf{a}_{i,\text{int}} = -G\sum_{\substack{j=1 \\ j\neq i}}^{N} m_j \frac{\mathbf{x}_i - \mathbf{x}_j}{\left(\|\mathbf{x}_i - \mathbf{x}_j\|^2 + \epsilon^2\right)^{3/2}} \tag{2.20}$$

The total acceleration is therefore the sum of the internal contribution and the galactic contribution defined in Section 2.3.1:

$$\mathbf{a}_i = \mathbf{a}_{i,\text{int}} + \mathbf{a}_{\text{gal}}(\mathbf{x}_i) \tag{2.21}$$

This formulation makes it possible to describe simultaneously the internal dynamics of the cluster and its response to the external gravitational field. Mutual interactions determine the evolution of the stellar structure, while the galactic potential influences the overall motion of the cluster and can promote the loss of stars that are no longer gravitationally bound to its internal component [3].

2.3.3 Initial Orbital Parameters

The initial orbit of the stellar cluster is defined through the semi-major axis a and the eccentricity e. These parameters determine the maximum distance R_{apo} and the minimum distance R_{peri} from the galactic centre:

$$R_{\text{apo}} = a(1+e) \qquad R_{\text{peri}} = a(1-e) \tag{2.22}$$

The cluster is initially placed at a distance R_{apo}, with a velocity perpendicular to the radial direction. Its magnitude is determined using the conservation of specific mechanical energy and specific angular momentum [9]:

$$E = \frac{v^2}{2} + \Phi_{\text{gal}}(R) \qquad L = Rv_{\text{t}} \tag{2.23}$$

where R is the distance from the galactic centre, v is the velocity magnitude and v_{t} is the tangential component. At apocentre and pericentre, the velocity is entirely tangential. Equating the energy at the two points and imposing conservation of angular momentum gives:

$$L^2 = \frac{2\left[\Phi_{\text{gal}}(R_{\text{apo}}) - \Phi_{\text{gal}}(R_{\text{peri}})\right]}{R_{\text{peri}}^{-2} - R_{\text{apo}}^{-2}} \tag{2.24}$$

The initial velocity at apocentre is therefore:

$$v_{\text{apo}} = \frac{L}{R_{\text{apo}}} \tag{2.25}$$

The velocity calculated in this way is assigned to all stars in the cluster, superimposed on their individual motions. Figure 2.5 shows, using OVITO [32], the evolution of the cluster for two different orbital eccentricities.

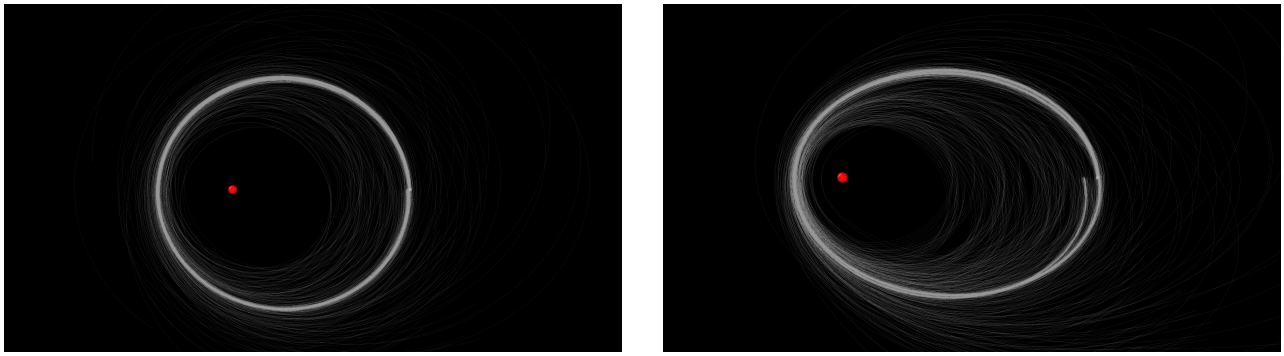


Figure 2.5: Evolution of the same stellar cluster after two orbits, with eccentricity $e = 0.3$ (left) and $e = 0.6$ (right). The white trajectories trace the movement of the stars, and the red dot indicates the galactic centre.
($N = 400$, $A = 1.0$, $M_{\text{gal}} = 2000.0$, $A_{\text{gal}} = 10.0$, $G = 1.0$, $\epsilon = 0.01$, $a = 300.0$, $\Delta t = 0.005$, *leapfrog*)

2.3.4 Numerical Integration

The evolution of the system is calculated using the *leapfrog* numerical integrator, a second-order symplectic method commonly used in long-term dynamical simulations [23]. Letting Δt denote the time

step, the method updates positions and velocities in three stages: first, it advances the stars by half a time step using their current velocities; it then updates the velocities by one full time step using the accelerations calculated at the intermediate positions; finally, it completes the displacement of the stars with a further half-step. The update equations are:

$$\mathbf{x}_i^{n+1/2} = \mathbf{x}_i^n + \frac{\Delta t}{2}\mathbf{v}_i^n \tag{2.26}$$

$$\mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \Delta t\, \mathbf{a}_i^{n+1/2} \tag{2.27}$$

$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^{n+1/2} + \frac{\Delta t}{2}\mathbf{v}_i^{n+1} \tag{2.28}$$

In this work, a time step of $\Delta t = 0.005$ is adopted for all simulations, which are run until the final time $T_{\text{end}} = 700.0$.
To verify the behaviour of the numerical integration, the relative variation of the total energy with respect to its initial value is monitored. Letting $E(t)$ denote the total energy of the system at time t and $E_0 = E(0)$ its initial value, this variation is defined as:

$$\delta_E(t) = \frac{\Delta E(t)}{E_0} = \frac{E(t) - E(0)}{E(0)} \tag{2.29}$$

This quantity is dimensionless and makes it possible to evaluate the relative deviation of the energy from its initial value. As an indicative reference, a maximum deviation of $\max_t |\delta_E(t)| \lesssim 10^{-3}$ can be considered small; in the case shown in Figure 2.6, the deviation takes values between approximately -5×10^{-3} and 1.25×10^{-2}, which are considered acceptable.

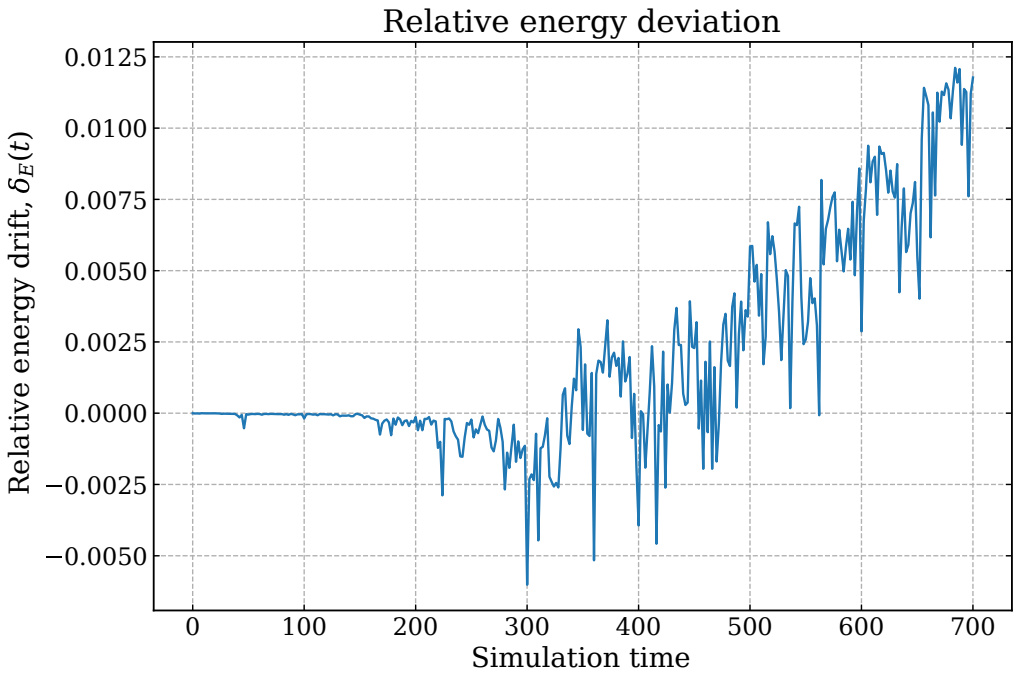


Figure 2.6: Relative energy drift $\delta_E(t)$ as a function of simulation time.
($N = 400$, $A = 1.0$, $M_{\text{gal}} = 2000.0$, $A_{\text{gal}} = 10.0$, $G = 1.0$, $\epsilon = 0.01$, $a = 300.0$, $e = 0.6$, $\Delta t = 0.005$, $T_{\text{end}} = 700.0$, *leapfrog*)

2.3.5 Identifying Stars Bound to the Cluster

A star is considered gravitationally bound to the cluster if its specific mechanical energy is negative. We therefore define the set of bound stars S_{bound} as:

$$S_{\text{bound}} = \{i \mid E_i < 0\}, \qquad E_i = K_i + \Phi_i \tag{2.30}$$

where K_i and Φ_i are, respectively, the specific kinetic energy and the specific gravitational potential of star i. The specific kinetic energy is calculated with respect to the velocity of the centre of mass of the bound component:

$$K_i = \frac{1}{2} \left\| \mathbf{v}_i - \mathbf{v}_{\text{CM,bound}} \right\|^2 \tag{2.31}$$

where $\mathbf{v}_i$ is the velocity of star i and $\mathbf{v}_{\text{CM,bound}}$ is the velocity of the centre of mass of the bound stars, defined in Section 2.3.6. The specific gravitational potential of star i is calculated by summing the contributions from all the other stars belonging to S_{bound}:

$$\Phi_i = -G \sum_{\substack{j \in S_{\text{bound}} \\ j \neq i}} \frac{m_j}{\left\| \mathbf{x}_i - \mathbf{x}_j \right\|} \tag{2.32}$$

where G is the universal gravitational constant, m_j is the mass of star j, and $\mathbf{x}_i$ and $\mathbf{x}_j$ are the positions of the stars. The calculation of the specific energy therefore depends on the set S_{bound}, which determines both the centre-of-mass velocity used as the reference for calculating the kinetic energy and the stars contributing to the gravitational potential. To resolve this mutual dependence, the selection is performed iteratively, updating S_{bound} and the velocity of its centre of mass at each step:

1. The star with the lowest gravitational potential is identified, and the eleven nearest stars are selected. Their mass-weighted mean velocity provides an initial estimate of the centre-of-mass velocity.

2. S_{bound} is initialised with all the stars in the cluster, and the specific energies are calculated. The stars with negative energy form the new set S_{bound}.

3. The energies are recalculated using the updated centre-of-mass velocity and the potential generated only by the stars belonging to the new set S_{bound}. The procedure is repeated until S_{bound} no longer changes, for a maximum of ten iterations. If at most three stars remain, no bound component is identified.

2.3.6 Centre of Mass of the Stars Bound to the Cluster

After identifying the set S_{bound} of bound stars, according to the procedure described in Section 2.3.5, the position and velocity of the centre of mass of the bound component are calculated as mass-weighted averages:

$$\mathbf{x}_{\text{CM,bound}} = \frac{\sum_{i \in S_{\text{bound}}} m_i \mathbf{x}_i}{\sum_{i \in S_{\text{bound}}} m_i} \tag{2.33}$$

The centre-of-mass velocity is calculated in the same way:

$$\mathbf{v}_{\text{CM,bound}} = \frac{\sum_{i \in S_{\text{bound}}} m_i \mathbf{v}_i}{\sum_{i \in S_{\text{bound}}} m_i} \tag{2.34}$$

In these expressions, m_i, $\mathbf{x}_i$ and $\mathbf{v}_i$ denote, respectively, the mass, position and velocity of star i.

2.3.7 Bound Mass Fraction

The bound mass fraction expresses the portion of the total mass that remains gravitationally bound to the cluster. Letting m_i denote the mass of star i, the bound mass is:

$$M_{\text{bound}} = \sum_{i \in S_{\text{bound}}} m_i \tag{2.35}$$

The bound mass fraction is obtained by dividing this quantity by the total mass M_{tot} of all the stars:

$$f_{\text{bound}} = \frac{M_{\text{bound}}}{M_{\text{tot}}} \tag{2.36}$$

The value $f_{\text{bound}} = 1.0$ corresponds to 100% of the initial mass still gravitationally bound to the cluster, whereas a value of zero indicates that no bound mass remains. The bound mass fraction f_{bound} is therefore a suitable quantity for describing the evolution of the gravitationally bound mass of the stellar cluster over time.

2.3.8 Half-Mass Radius

The half-mass radius indicates the distance from the centre of mass within which half of the bound mass is contained. For each star belonging to S_{bound}, the distance from the centre of mass is calculated:

$$d_i = \|\mathbf{x}_i - \mathbf{x}_{\text{CM,bound}}\| \qquad \forall i \in S_{\text{bound}} \tag{2.37}$$

The stars are then sorted by increasing distance, and the cumulative mass of the bound component is calculated. The mass contained within a distance r, denoted by $M_{\text{bound}}(< r)$, is:

$$M_{\text{bound}}(< r) = \sum_{\substack{i \in S_{\text{bound}} \\ d_i \leq r}} m_i \tag{2.38}$$

The half-mass radius is the minimum distance at which the cumulative mass reaches at least half of the bound mass:

$$r_{1/2} = \min \left\{ r : M_{\text{bound}}(< r) \geq \frac{M_{\text{bound}}}{2} \right\} \tag{2.39}$$

If no gravitationally bound component is identified, the half-mass radius is set to zero. The half-mass radius $r_{1/2}$ is therefore a suitable quantity for describing the evolution of the spatial distribution of the bound mass over time.

Chapter 3

Dataset and Machine Learning

The construction of the *dataset* begins with the definition of the star cluster parameters, the galactic potential and the orbits, as well as the numerical integration conditions. The results are then analyzed through the distribution of the final quantities to study the relationship between the orbital parameters and the evolution of the clusters. Based on this analysis, a binary classification problem is defined by establishing the criteria for distinguishing surviving clusters from dissolved ones, the input variables, and the division of the *dataset* for training and evaluation. The *F1-score*, used to measure classifier performance, is also introduced. Finally, the *Support Vector Classifier* (SVC) and *Multi-Layer Perceptron* (MLP) classifiers are presented, describing how they work and comparing their results.

3.1 Dataset Generation

The simulations are performed by varying the orbital parameters of the cluster, namely the semi-major axis and eccentricity. The results obtained constitute the *dataset* used to study how different orbital conditions influence the evolution of star clusters.

3.1.1 Star Cluster Parameters

The star cluster is initialized according to the Plummer model, described in Section 2.2.1. The number of stars, $N = 200$, the scale parameter $A = 1.0$, and the total mass $M_{\text{tot}} = 1.0$ are kept fixed in all simulations. All stars have the same mass, calculated as:

$$m = \frac{M_{\text{tot}}}{N} = 0.005 \tag{3.1}$$

To obtain different initial realizations of the cluster, 15 distinct random seeds are used, producing different initial distributions of stellar positions and velocities. This makes it possible to evaluate the influence of initial conditions on the results, avoiding basing the analysis on a single realization of the cluster.

3.1.2 Galactic and Orbital Parameters

The external gravitational potential is described by the Plummer model, introduced in Section 2.3.1, with a galactic mass of $M_{\text{gal}} = 3000.0$ and a scale parameter of $A_{\text{gal}} = 20.0$, both kept constant across all simulations. The orbits are defined by the semi-major axis a and the eccentricity e. The former takes 30 values between 10.0 and 300.0, in increments of 10.0:

$$a \in \{10, 20, 30, \ldots, 290, 300\}$$

The eccentricity takes 38 values between 0.0 and 0.925, in increments of 0.025:

$$e \in \{0, 0.025, 0.050, \ldots, 0.900, 0.925\}$$

Considering all combinations of the two parameters yields $30 \times 38 = 1140$ orbital configurations. Each configuration is simulated using the 15 initial cluster realizations described in Section 3.1.1. The *dataset* therefore comprises a total of $15 \times 1140 = 17100$ simulations.

3.1.3 Integration Parameters

The simulations use a gravitational constant of $G = 1.0$, which allows all quantities to be expressed in a consistent system of units, avoiding conversions between different units of measurement. Numerical integration is performed using the *leapfrog* method [12], with a time step of $\Delta t = 0.005$. The softening parameter ϵ, introduced in Section 2.3.2, is set to 0.01.
The final simulation time is set to $T_{\text{end}} = 700.0$, in order to limit the influence of cluster dissolution caused by its internal dynamical evolution and focus on the effects of galactic tidal stripping [4].

3.1.4 Summary of Parameters

Table 3.1 summarizes the parameters adopted for the simulations, distinguishing those kept constant from those varied to explore the different orbital configurations.

Table 3.1: Parameters adopted for dataset generation.

Parameter	**Symbol**	**Value**
Star Cluster		
Number of stars	N	200
Total mass	M_{tot}	1.0
Mass of a star	m	0.005
Plummer model scale parameter	A	1.0
Number of random seeds used	—	15
Galactic Potential		
Galactic mass	M_{gal}	3000.0
Galactic scale parameter	A_{gal}	20.0
Orbital Parameters		
Semi-major axis	a	[10.0, 300.0], step 10.0
Eccentricity	e	[0.0, 0.925], step 0.025
Numerical Integration		
Gravitational constant	G	1.0
Time step	Δt	0.005
Softening parameter	ϵ	0.01
Final time	T_{end}	700.0
Integrator	—	*leapfrog*

3.2 Dataset Analysis

3.2.1 Distribution of Final Quantities

To analyze the simulation results, we consider the histograms of the final bound mass fraction f_{bound} and the ratio between the final half-mass radius $r_{1/2}$ and its initial value $r_{1/2,0}$, defined as:

$$\mathcal{R}_{1/2} = \frac{r_{1/2}}{r_{1/2,0}} \tag{3.2}$$

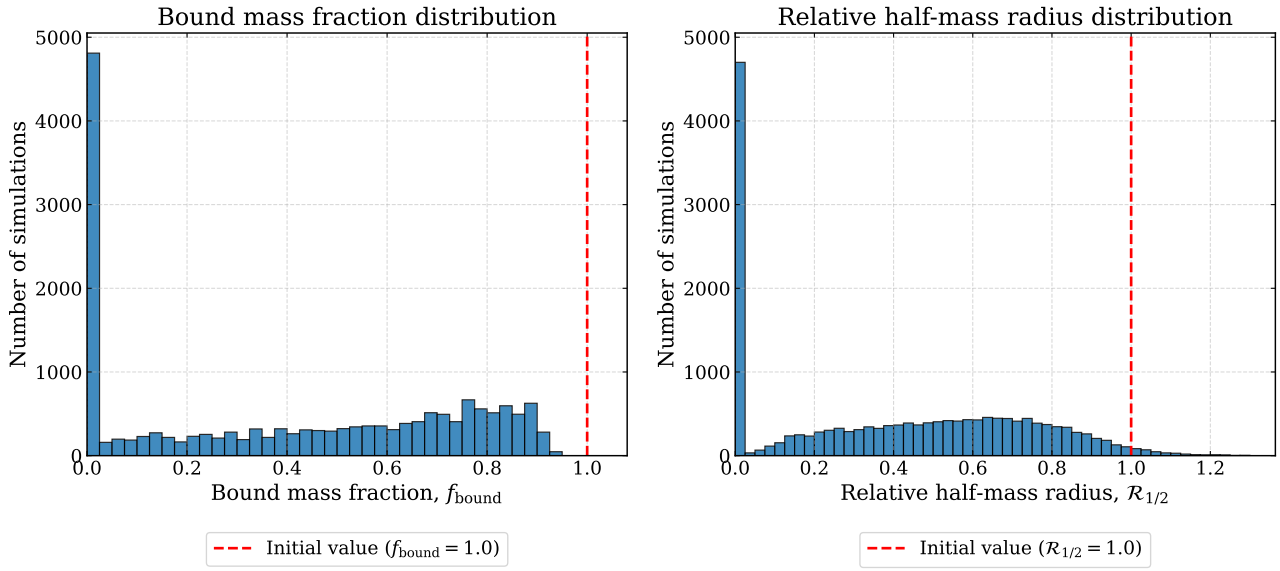


Figure 3.1: Distribution of the final bound mass fraction f_{bound} (left) and the relative half-mass radius $\mathcal{R}_{1/2}$ (right).

Figure 3.1 shows that both distributions exhibit a bimodal pattern, characterized by two distinct groups of values. In 4689 simulations, corresponding to 27.42% of the *dataset*, both the bound mass fraction f_{bound} and the relative half-mass radius $\mathcal{R}_{1/2}$ are zero. In these cases, no stars remain gravitationally bound to the cluster, and the stars have become completely dispersed under the effect of galactic tidal forces.
Outside the peak at zero, both distributions exhibit a flatter pattern, with values spread over a fairly wide range. However, most clusters have a bound mass fraction below 1.0 and a relative half-mass radius also below 1.0.
This indicates that, in most cases, the clusters have lost part of their mass and contracted relative to their initial configuration.
The two distributions are therefore consistent with each other, showing how mass loss is generally accompanied by a reduction in the cluster's spatial extent.

3.2.2 Relationship Between Orbital Parameters and Simulation Outcomes

Figure 3.2 shows the distributions of the final bound mass fraction f_{bound} and the relative half-mass radius $\mathcal{R}_{1/2}$, as a function of the relative semi-major axis $\tilde{a}$ and the eccentricity e. The relative semi-major axis is defined as $\tilde{a} = \frac{a}{A_{\text{gal}}}$ and is a dimensionless quantity expressing the orbital extent in units of the galactic scale parameter A_{gal}. Each cell represents the mean of the results from the 15 initial realizations for a given orbital configuration.

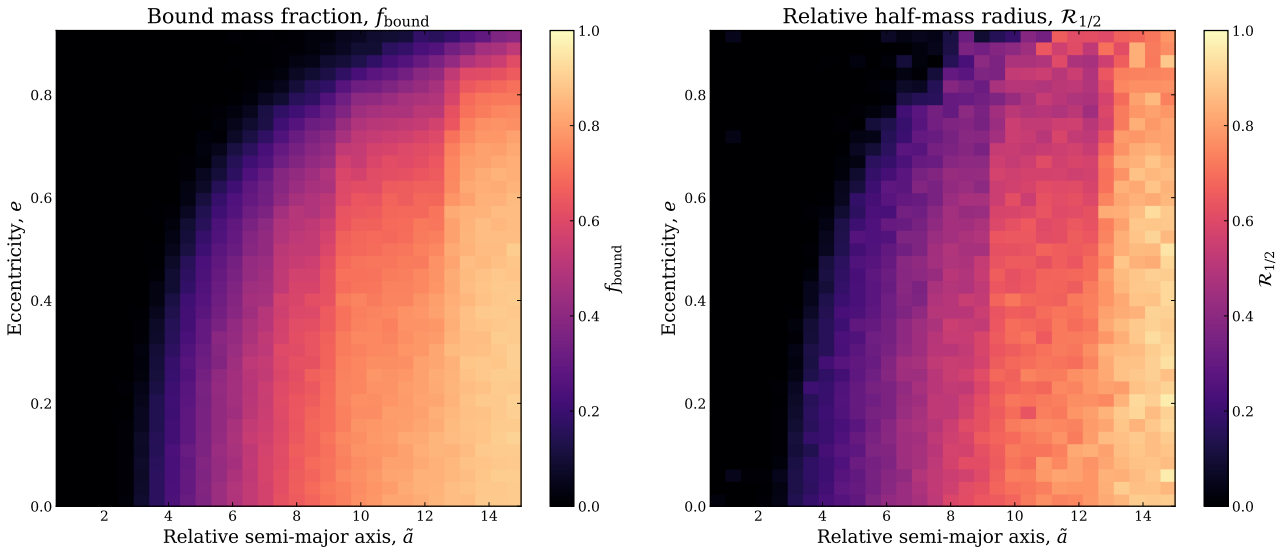


Figure 3.2: Distribution of the mean values of the final bound mass fraction f_{bound} (left) and the relative half-mass radius $\mathcal{R}_{1/2}$ (right), as a function of the relative semi-major axis $\tilde{a}$ and the eccentricity e.

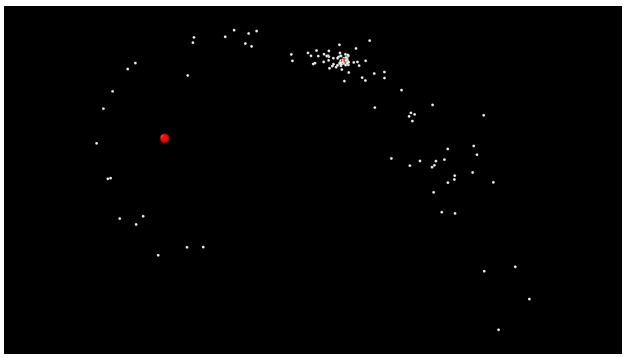
The distributions reveal a clearly identifiable transition line between orbital configurations for which the bound mass fraction and the relative half-mass radius are zero or close to zero, and those in which both quantities remain positive. This separation provides further evidence of the relationship between the values of the quantities describing the cluster's state and its evolution.
In particular, for relative semi-major axis values below approximately 3.0, both quantities tend to approach zero, indicating the complete dissolution of the cluster due to the greater intensity of tidal effects at smaller distances from the galactic center.
Eccentricity also affects the evolution of the cluster: for the same semi-major axis, higher values result in a smaller distance from the galactic center at pericenter, exposing the cluster to stronger tidal effects and promoting its dissolution.

3.3 Definition of the Classification Problem

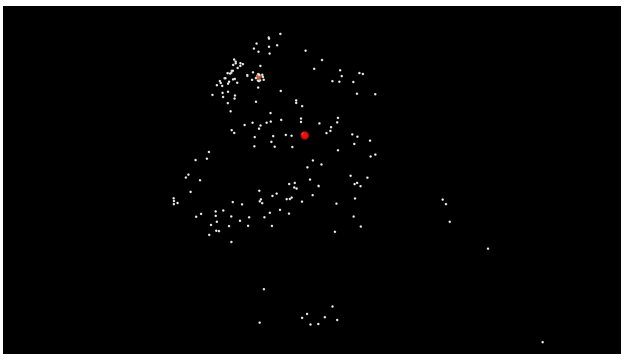
To apply machine learning methods, the *dataset* is organized into a set of input variables and a set of labels, defining the criteria for assigning classes and the subsequent division of the data for model training and evaluation.

3.3.1 Classification of Star Clusters

As shown in Section 3.2.1, at the end of the simulation star clusters can exhibit different evolutionary outcomes: some retain a compact, gravitationally bound structure, while others undergo mass loss and changes in spatial extent that significantly alter their initial structure. These differences make it possible to distinguish two classes of evolutionary outcome: surviving clusters and dissolved clusters. In this work, this distinction is used to formulate a binary classification problem, in which each cluster is assigned a label corresponding to its class.



Surviving star cluster
($a = 300.0$, $e = 0.5$)
($f_{\text{bound}} = 0.93$, $\mathcal{R}_{1/2} = 1.002$)



Dissolved star cluster
($a = 140.0$, $e = 0.8$)
($f_{\text{bound}} = 0.065$, $\mathcal{R}_{1/2} = 0.246$)

Figure 3.3: Evolution of the same star cluster, generated from the same initial configuration and subjected to different orbital conditions. The images show the configurations reached at time T_{end}: in one case the cluster survives, while in the other it dissolves. The configurations are visualized using OVITO [32]: stars are shown in white, the galactic center in red, and the center of mass of the bound stars in orange.
($N = 200$, $A = 1.0$, $M_{\text{gal}} = 3000.0$, $A_{\text{gal}} = 20.0$, $G = 1.0$, $\epsilon = 0.01$, $\Delta t = 0.005$, $T_{\text{end}} = 700.0$, *leapfrog*)

3.3.2 Formal Definition of the Sets

Let $\mathcal{N}$ be the set of all star clusters i. As described in Section 3.3.1, the clusters are divided into two subsets, $\mathcal{N}_{\text{survivors}}$ and $\mathcal{N}_{\text{dissolved}}$, containing the surviving and dissolved clusters, respectively. This division is based on two quantities associated with each cluster i: the bound mass fraction $f_{\text{bound},i}$ and the ratio between the final and initial half-mass radii $\mathcal{R}_{1/2,i}$. The adopted threshold values are 0.1 for the bound mass fraction and 0.2 for the relative half-mass radius. These thresholds were identified through an analysis of the cluster configurations in the initial and final stages of the simulations, as well as of the distributions of the final quantities presented in Sections 3.2.1 and 3.2.2. The sets $\mathcal{N}_{\text{survivors}}$ and $\mathcal{N}_{\text{dissolved}}$ are defined as follows:

$$\mathcal{N}_{\text{survivors}} = \left\{ i \in \mathcal{N} \,\middle|\, f_{\text{bound},i} > 0.1 \ \wedge \ \mathcal{R}_{1/2,i} > 0.2 \right\}$$

$$\mathcal{N}_{\text{dissolved}} = \left\{ i \in \mathcal{N} \,\middle|\, f_{\text{bound},i} \leq 0.1 \ \vee \ \mathcal{R}_{1/2,i} \leq 0.2 \right\}$$

The two sets are disjoint and exhaustive with respect to $\mathcal{N}$, since they satisfy the following properties:

$$\mathcal{N}_{\text{survivors}} \cap \mathcal{N}_{\text{dissolved}} = \emptyset \qquad \mathcal{N}_{\text{survivors}} \cup \mathcal{N}_{\text{dissolved}} = \mathcal{N}.$$

Out of a total of $|\mathcal{N}| = 17\,100$ analyzed simulations, the set $\mathcal{N}_{\text{survivors}}$ contains 11 695 clusters, corresponding to 68.39% of the total, while $\mathcal{N}_{\text{dissolved}}$ contains 5 405 clusters, corresponding to 31.61%.

3.3.3 Input and Output Variables

For each simulation, the input variable vector is defined as

$$\mathbf{x} = \begin{bmatrix} a \\ e \end{bmatrix} \tag{3.3}$$

These parameters describe the initial orbital conditions and constitute the information available to the model for predicting the simulation outcome. Considering all simulations, the input matrix is represented by

$$X \in \mathbb{R}^{|\mathcal{N}| \times 2} \tag{3.4}$$

while the labels associated with the simulations form the vector

$$Y \in \{0,1\}^{|\mathcal{N}|} \tag{3.5}$$

Using the sets $\mathcal{N}_{\text{survivors}}$ and $\mathcal{N}_{\text{dissolved}}$, defined in Section 3.3.2, each simulation i is assigned a label according to the following definition:

$$Y_i = \begin{cases} 1 & \text{if } i \in \mathcal{N}_{\text{survivors}} \\ 0 & \text{if } i \in \mathcal{N}_{\text{dissolved}} \end{cases} \tag{3.6}$$

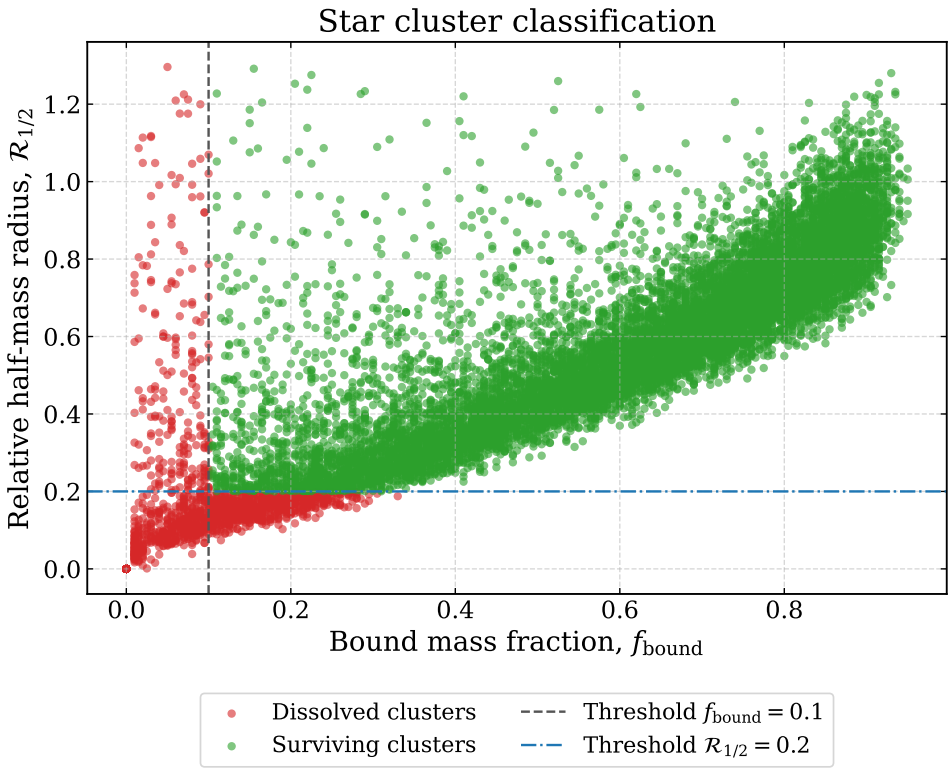


Figure 3.4: Outcome of the binary classification of star clusters.

3.3.4 Training and Test Set Split

To evaluate the models' ability to generalize, the *dataset* is randomly divided into a training set containing 80% of the simulations and a test set containing the remaining 20%. Thus, out of a total of 17100 simulations, there are 13680 training samples and 3420 evaluation samples. The split is stratified by labels, so as to maintain comparable proportions of surviving and dissolved clusters in the two sets. Sample selection is performed using a pseudo-random procedure controlled by a predefined seed, kept identical for both models. This ensures that the two splits are the same and that the results are comparable and reproducible. The *training set* is used to learn the relationship between the orbital parameters and the labels, while the *test set* is used exclusively to evaluate model performance on simulations not used during training.

3.3.5 Comparison Using the F1-score

To evaluate model performance, precision and recall are considered. These are two complementary measures that describe, respectively, the reliability of predictions and the ability to correctly identify elements belonging to a given class. Both are calculated separately for surviving and dissolved clusters. Let TP denote the number of elements belonging to the considered class that are correctly classified, FP the number of elements belonging to the other class but incorrectly assigned to the considered

class, and FN the number of elements belonging to the considered class but incorrectly assigned to the other class. The resulting measures are:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.7}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.8}$$

To summarize both measures in a single value, the *F1-score* is used, defined as their harmonic mean [29]:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.9}$$

The *F1-score* is therefore calculated separately for each class and takes values between 0 and 1: values closer to 1 indicate a more accurate classification. Finally, to obtain an overall measure of performance, the weighted average of the *F1-scores* of the two classes is used, with each contribution proportional to the number of simulations belonging to the respective class.

3.4 Application of the Models

In this section, two machine learning models are applied to classify star clusters based on their initial orbital parameters. The models considered are a *Support Vector Classifier* and a *Multi-Layer Perceptron* neural network, whose main characteristics are described. Both are trained using the *training set* data and subsequently evaluated on the *test set*, comparing their performance.

3.4.1 Feature Standardization

Before training, the input variables are standardized to make them comparable, preventing differences in their numerical ranges from influencing the learning process.
The transformation applied to each variable is:

$$x' = \frac{x - \mu}{\sigma} \tag{3.10}$$

where x is the original value, while μ and σ denote the mean and standard deviation, respectively. These parameters are calculated exclusively on the *training set* and subsequently used to transform the *test set* as well. This prevents information contained in the test data from indirectly influencing model training [15].

3.4.2 Support Vector Classifier

Model

The first classifier used is the *Support Vector Classifier* (SVC), based on *Support Vector Machines* (SVM) [6]. Its objective is to identify a decision boundary that separates surviving clusters from dissolved ones in the space defined by the orbital parameters. The position of this boundary depends primarily on the samples closest to it, called support vectors, which determine its shape. To allow for non-linear separation, a second-degree polynomial kernel is adopted, defined as:

$$K(\mathbf{x}, \mathbf{z}) = (\gamma\, \mathbf{x} \cdot \mathbf{z} + r)^2 \tag{3.11}$$

A kernel makes it possible to evaluate the relationship between two samples through a mathematical function, allowing decision boundaries to be identified even when the classes cannot be separated by a simple straight line [27]. In this expression, $\mathbf{x}$ and $\mathbf{z}$ represent two vectors containing the standardized orbital parameters, while $\mathbf{x} \cdot \mathbf{z}$ denotes their dot product.
The parameter γ determines the scale of the dot product and is calculated according to the following

relation [33]:

$$\gamma = \frac{1}{n_{\text{feature}} \cdot \text{Var}(X)} \tag{3.12}$$

where n_{feature} denotes the number of input variables and $\text{Var}(X)$ their average variance. Since the two variables have been standardized (Section 3.4.1), their variance is equal to one. Therefore:

$$\gamma = \frac{1}{2 \cdot 1} = 0.5$$

The parameter r, corresponding to the constant term of the kernel, is set to 1.0. This choice makes it possible to include, in addition to quadratic terms and interactions between the variables, linear terms and a constant term. Substituting this value into Equation 3.11 gives:

$$K(\mathbf{x}, \mathbf{z}) = \gamma^2(\mathbf{x} \cdot \mathbf{z})^2 + 2\gamma(\mathbf{x} \cdot \mathbf{z}) + 1 \tag{3.13}$$

Results

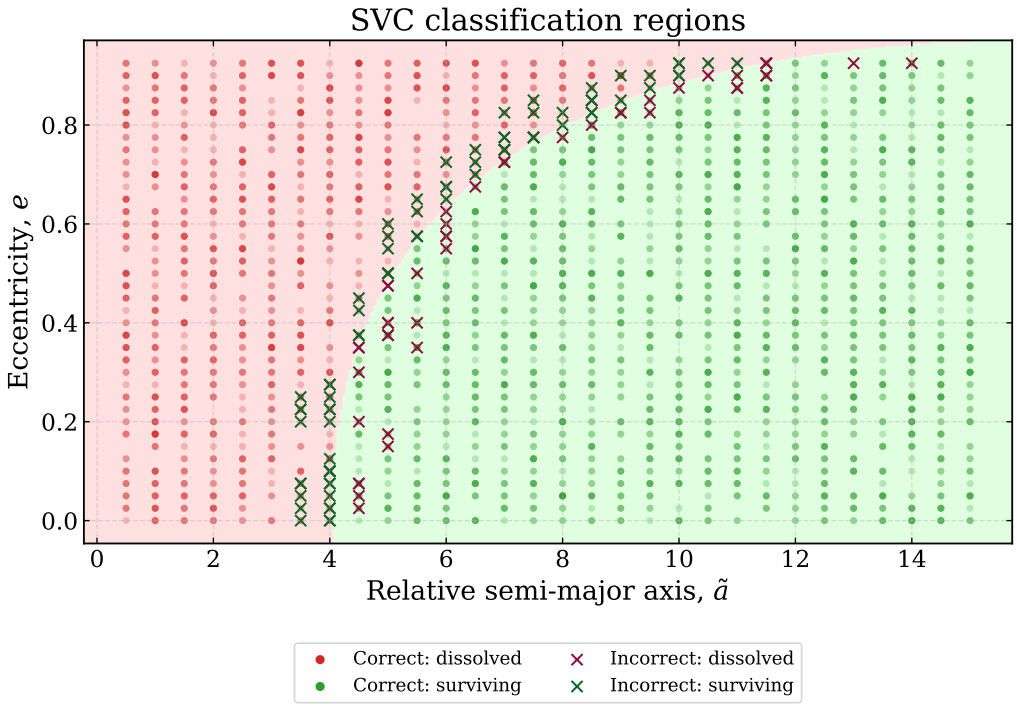


Figure 3.5: Decision regions of the SVC classifier as a function of the relative semi-major axis $\tilde{a}$ and eccentricity e. The background represents the model predictions, while the points show the *test set* simulations; crosses indicate misclassifications. (*F1-score* = 0.9644)

On the *test set*, the classifier achieves a weighted *F1-score* of 0.9644. Performance is high for both classes, with a value of 0.95 for dissolved clusters and 0.97 for surviving clusters. Thanks to the expanded form of the kernel, described in Equation 3.13, the decision boundary can be expressed through an explicit quadratic equation.
Expressing the coefficients in the original orbital parameter space gives:

$$f(a, e) = -0.0001\,a^2 - 9.1336\,e^2 - 0.0747\,ae + 0.1614\,a + 6.4634\,e - 12.1809 \tag{3.14}$$

Considering the coefficients alone, they might appear to indicate a greater influence of eccentricity on the classification, since the terms containing it have larger absolute values. However, it must be taken into account that the semi-major axis assumes numerically much larger values than eccentricity. The different numerical scales of the two parameters are therefore reflected in the coefficients learned

by the model, which has identified a relationship involving both parameters and their combination in determining the classification outcome.

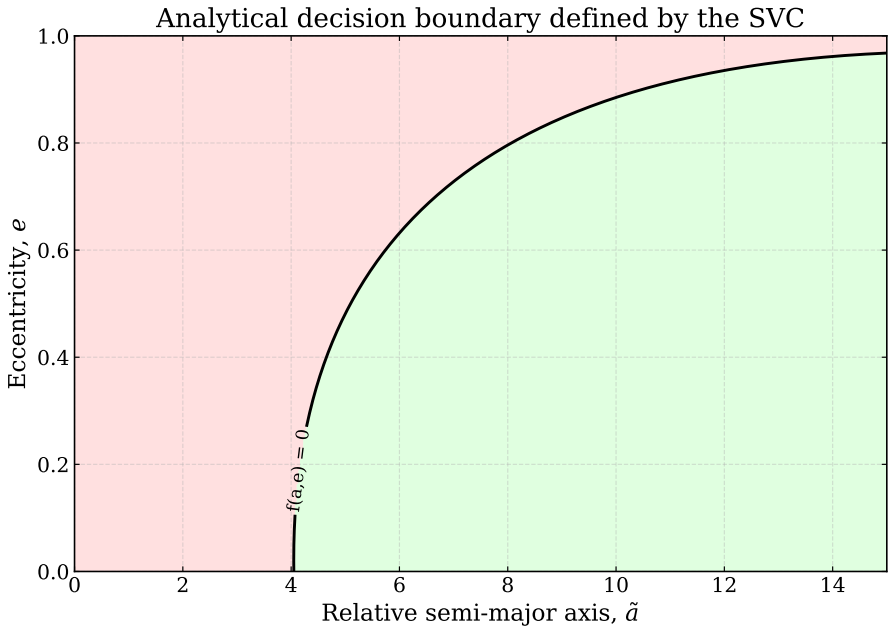


Figure 3.6: Graphical representation of the decision boundary defined by Equation 3.14.
The green region represents the cluster survival zone ($f(a, e) > 0$), while the red region represents the dissolution zone ($f(a, e) < 0$).

3.4.3 Multi-Layer Perceptron

Model

The second classifier used is a *Multi-Layer Perceptron* (MLP), an artificial neural network composed of neurons arranged in different layers [10]. Each neuron receives the values produced by the previous layer, combines them using numerical weights, and returns an output that is passed to the subsequent neurons. During training, these weights are adjusted to improve the model's ability to correctly classify the clusters [26]. The adopted architecture comprises two hidden layers, containing 16 and 8 neurons, respectively, and an output layer for binary classification. The hidden layers use the ReLU (*Rectified Linear Unit*) activation function, defined as:

$$\mathrm{ReLU}(x) = \max(0, x)$$

This function returns zero for negative values and leaves positive values unchanged. Its introduction allows the network to learn non-linear relationships between the input variables [8].
Weight updates during training are performed by the Adam algorithm, an optimization method that automatically adjusts the magnitude of the updates based on the errors made by the network and the behavior of the gradients across iterations [16].
Training consists of a maximum of 500 iterations.

Results

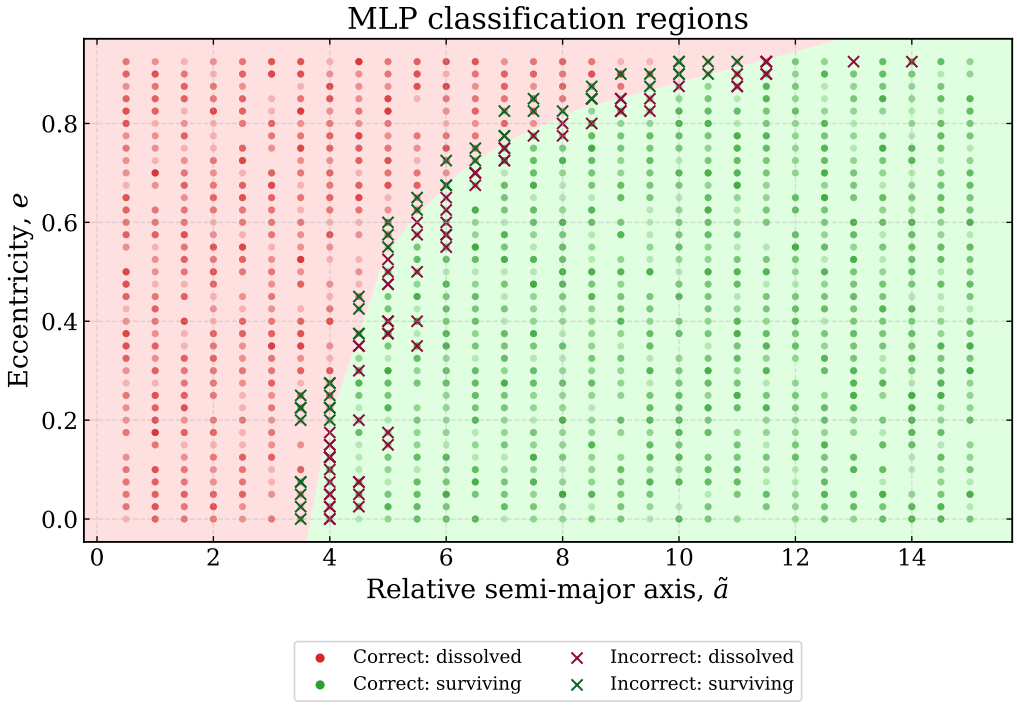


Figure 3.7: Decision regions of the MLP classifier as a function of the relative semi-major axis $\tilde{a}$ and eccentricity e. The background represents the model predictions, while the points show the *test set* simulations; crosses indicate misclassifications. (*F1-score* = 0.9642)

On the *test set*, the classifier achieves a weighted *F1-score* of 0.9642. Considering the two classes separately, the value is 0.95 for dissolved clusters and 0.97 for surviving clusters. Unlike the SVC (Section 3.4.2), the MLP decision boundary cannot be described by a simple quadratic equation, since it depends on the weights and activation functions of the different neurons. However, the decision boundaries of the two classifiers can be compared directly, as shown in Figure 3.8.

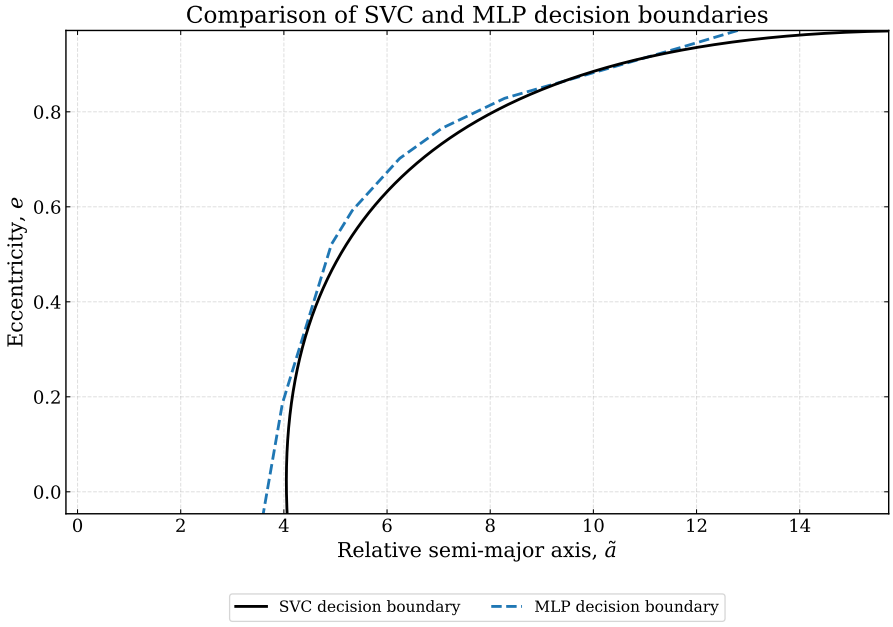


Figure 3.8: Comparison between the decision boundaries of the SVC (solid line) and MLP (dashed line) classifiers.

Chapter 4

Software Implementation

The simulation implementation was developed using the Python programming language [34] and the REBOUND library [24], version 5.0.0, for the numerical integration of gravitational N-body systems. REBOUND provides the tools required to manage particles and integrate their respective equations of motion. NumPy (2.0.2) [13] and SciPy (1.18.0) [35] were used for numerical operations, while Numba (0.60.0) [19] was employed to accelerate the computation of galactic accelerations through *just-in-time* compilation.
Project-specific features, including the generation of clusters according to the Plummer model, the definition of the galactic potential, and the calculation of diagnostic quantities, were implemented using dedicated modules. Their main functionalities are summarized in Table 4.1.

Table 4.1: Software modules developed and their respective functionalities.

Module	**Functionality**
`plummer.py`	Generation of initial positions and velocities and construction of star clusters.
`galaxy.py`	Definition of the galactic potential, calculation of the gravitational acceleration exerted by the galaxy on the stars, and determination of the initial orbital conditions.
`simulation.py`	Interface with REBOUND and management of stars, integration parameters, and dynamical evolution.
`cluster_generation.py`	Coordination of cluster generation and creation of the corresponding REBOUND simulation, integrating the functionalities of `plummer.py` and `simulation.py`.
`galactic_tidal_stripping.py`	Coordination of the galactic system configuration and orbital conditions with the creation and execution of the simulation, integrating the functionalities of `galaxy.py` and `cluster_generation.py`.
`cluster_diagnostics.py`, `galactic_diagnostics.py`	Calculation of energies, the virial ratio, bound stars, the bound mass fraction, and the half-mass radius.

At the end of the simulations, the dataset used for the analyses was generated in CSV (*Comma-Separated Values*) format [28]. The analyses were carried out separately from the simulator code, using Jupyter *notebooks* [17] running in the Google Colab *cloud* environment [11]. The runtime used provided NumPy (2.1.3), pandas (2.2.3) [20], and Matplotlib (3.10.0) [14] for numerical operations, data manipulation, and plot generation. The training and evaluation of the SVC and MLP classifiers were performed using scikit-learn (1.6.1) [21].
The source code, configuration files, *notebooks*, and execution instructions are available in the project's public GitHub repository, in the "Final public release" *release* [1], accessible at: `https://github.com/Qw3rty26/GTS_ML/`.

Chapter 5

Conclusions

Both classifiers achieved high and comparable performance. The SVC, however, has an advantage in terms of interpretability, as it allows the decision boundary to be expressed through an explicit quadratic equation. Based on the results, the decision boundaries of the two models appear to closely reproduce the separation that emerges from the distribution of outcomes presented in Section 3.2.2. Classification errors are concentrated mainly in the transition region between the two regions, where clusters characterized by the same orbital parameters can reach different outcomes depending on their initial pseudo-random realization. This variability seems to suggest that the semi-major axis and eccentricity, considered as the only input variables, are not sufficient to fully describe the evolution of star clusters.
To investigate this aspect further, it may be useful to increase the number of realizations for each orbital configuration and include additional diagnostic quantities among the input variables.

A further aspect that emerged during the study concerns cluster dissolution due to their internal dynamical evolution, which can occur even independently of the galactic potential. To limit its influence on the results, the final simulation time was set to $T_{\text{end}} = 700$. A possible direction for future work would be to increase the number of stars in each cluster, thereby reducing the effects associated with the finite number of particles and obtaining a more stable description of its internal evolution.

A positive result, on the other hand, concerns the Plummer model used to generate the initial cluster conditions. Although its theoretical description is formulated in the limit $N \to \infty$, it proved effective even for clusters containing only 200 stars. This result was supported by the rescaling of the initial velocities, which made it possible to compensate for deviations from virial equilibrium due to sampling with a finite number of stars.

Finally, the range of semi-major axes explored represents a further limitation of the *dataset*.
It would be interesting to extend this range and verify whether the models can correctly predict the outcomes of configurations characterized by orbital parameters outside the intervals used during training. This analysis would make it possible to evaluate the classifiers' ability to extrapolate the learned relationships and apply them to configurations that have never been simulated before.

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